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1. Review 2x2 analytic solution

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2x2 system

As a warm up, let's revisit the following 2×2 system which describes the flow of medicine betwee was modeled using the following system of ODEs:

$$\begin{aligned}\frac{dh_1}{dt} &= a(h_2 - h_1), \\ \frac{dh_2}{dt} &= a(h_1 - h_2).\end{aligned}$$

We then recast this system in matrix form as

$$\dot{\mathbf{x}} = \begin{bmatrix} -a & a \\ a & -a \end{bmatrix} \mathbf{x},$$

where $\mathbf{x} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$. Let's set $a = 1$, and use the initial conditions $\mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

We know that the solution can be written in the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2.$$

In a previous exercise, we found the c_i, λ_i and v_i by hand. This is much easier to do with MATLAB. construct the analytic solution to the above system of equations.

Script ?

 Save

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 [MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

```
1 % Input the 2x2 matrix describing your linear system as a variable A
2 A = [-1 1; 1 -1];
3 % Input your initial conditions as a column vector x0
4 x0 = [0; 1];
5 % Use eig(A) to find the eigenvalues and eigenvectors of A.
6 % Then define the eigenvectors as column vectors v1 and v2, and the eigenval
7 % so that A*v1 = lambda1*v1, etc.
8 [V,D] = eig(A);
9 v1 = V(:,1);
10 v2 = V(:,2);
11 lambda1 = D(1,1);
12 lambda2 = D(2,2);
13 % Calculate the column vector c = [c1;c2] from the initial conditions using
14 c = inv(V)*x0;
15 c1 = c(1,1); c2 = c(2,1);
16
```

 Run Script



Assessment: All Tests Passed

Submit ?

-  Value of c
-  Inv used
-  eig used
-  value of v1
-  value of v2
-

✔ value of lambda1

✔ value of lambda2

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